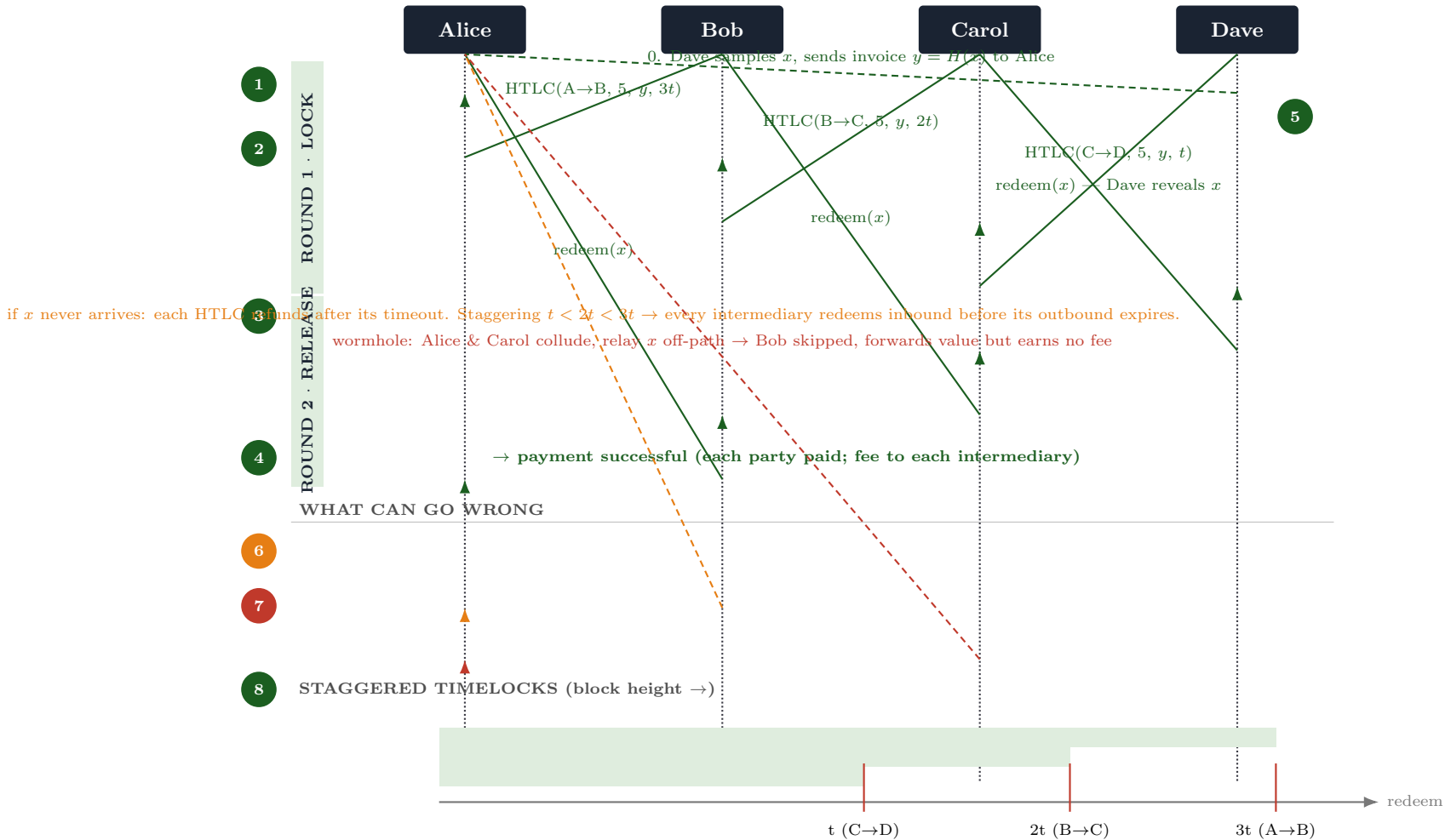


Multi-hop payment: Alice → Bob → Carol → Dave

annotated with multihop.spthy lemmas — see rail →



if x never arrives: each HTLC times out after its timeout. Staggering $t < 2t < 3t \rightarrow$ every intermediary redeems inbound before its outbound expires.
wormhole: Alice & Carol collude, relay x off-path $\rightarrow$ Bob skipped, forwards value but earns no fee

WHAT CAN GO WRONG

Rough draft — multihop.spthy lemmas mapped to the 4-party flow. Next: cltv_blocks / Clock / timeout.

LEMMAS · multihop.spthy

- 1 Invoice & preimage authentication
 - Invoice_Authenticates_Settlement ·
 - Preimage_Secret_Until_Released
 - Forged_Invoice_Requires_Key_Compromise ·
- 2 Lock causality (forward needs previous)
 - Forward1_Requires_Offer ·
 - Forward2_Requires_Forward1
 - HTLC_On_Opened_Channel ✓
- 3 Release causality (redeem needs preimage)
 - Fulfill_Requires_Forward2 ·
 - Claim_Requires_Release
 - Settle_Requires_Receiver_Release ✓
- 4 Atomicity + non-vacuity
 - Payment_Atomicity_Under_Liveness ✓
 - Multihop_Payment_Possible ✓ (witness)
- 5 Value & fee conservation
 - EndToEnd_Value_Conservation ·
 - Fee_Conservation_Hop1/2
 - Fee_Strictly_Positive_Hop1/2 ·
 - Receiver_Paid_Invoice_Amount ✓
- 6 Refund / grieving
 - Refund_Requires_Timeout ·
 - Loss_Requires_Inaction ✓
 - Honest_Intermediary_Refunds ✓ (reachable re-attack/DoS)
- 7 Wormhole fee theft
 - Wormhole_Fee_Theft_Reachable ✓
 - Wormhole_Steals_Exactly_The_Fees ✓ (theft = path fees)
- 8 Intermediary safety (under liveness T2b)
 - Intermediary_Never_Loses_Under_Liveness ✓
 - Timeout_Race_Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).
green = all-traces safety · orange = be-
nign/reachable
red = attack shown reachable (by design, no
PTLC).
Channel-layer lemmas
(open/update/close/punish)
live on the Alice↔Bob figure.